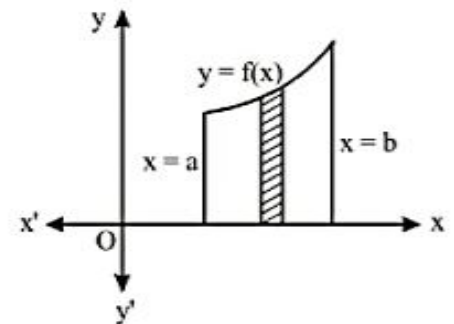


AREA UNDER THE CURVE

DIFFERENT CASES OF BOUNDED AREA :

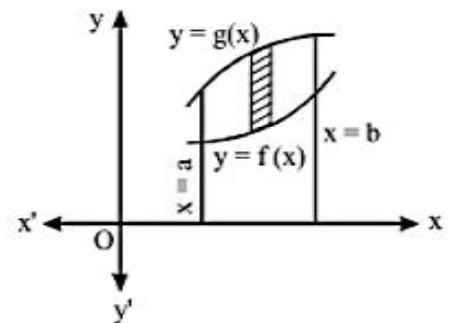
1. The area bounded by the continuous curve $y = f(x)$, the axis of x and the ordinates $x = a$ and $x = b$ (where $b > a$) is given by

$$A = \int_a^b f(x) dx = \int_a^b y dx$$



2. The area bounded by the straight line $x = a$, $x = b$ ($a < b$) and the curves $y = f(x)$ and $y = g(x)$, provided $f(x) < g(x)$ (where $a \leq x \leq b$), is given by

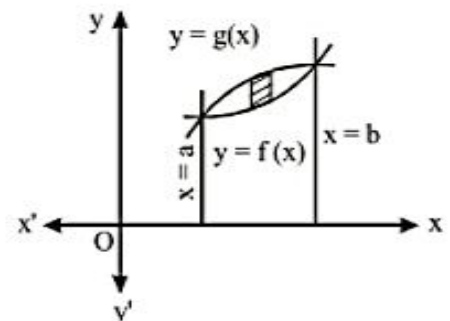
$$A = \int_a^b [g(x) - f(x)] dx$$



3. When two curves $y = f(x)$ and $y = g(x)$ intersect, the bounded area is

$$A = \int_a^b [g(x) - f(x)] dx ; \text{ where } a < b.$$

where a and b are the roots of the equation $f(x) = g(x)$.

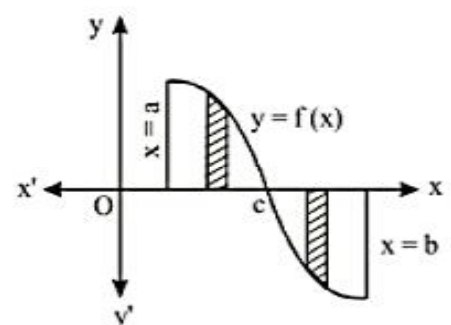


4. If some part of a curve lies below the x -axis, then its area becomes negative but area cannot be negative. Therefore, we take its modulus.

If the curves crosses the x -axis at c , then the area bounded by the curve $y = f(x)$ and ordinates $x = a$ and $x = b$

(where $b > a$) is given by $A = \left| \int_a^c f(x) dx \right| + \left| \int_c^b f(x) dx \right|$

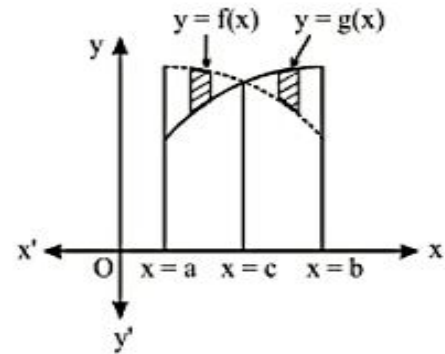
$$A = \int_a^c f(x) dx - \int_c^b f(x) dx$$



5. The area bounded by $y = f(x)$ and $y = g(x)$ (where $a \leq x \leq b$), when they intersect at $x = c \in (a, b)$ is given by

$$A = \int_a^b |f(x) - g(x)| dx$$

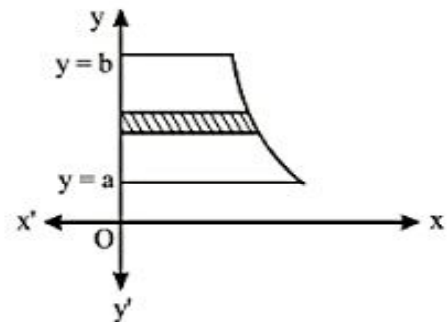
or
$$\int_a^c (f(x) - g(x)) dx + \int_c^b (g(x) - f(x)) dx$$



DIFFERENT CASES OF BOUNDED AREA :

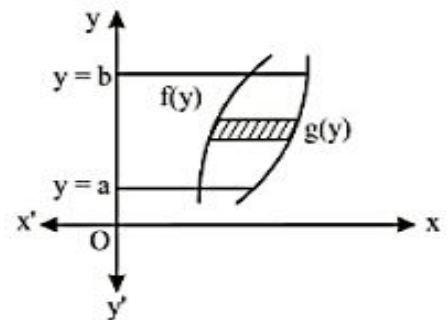
1. The area bounded by the continuous curve $x = f(y)$, the axis of y and the abscissa $y = a$ and $y = b$ (where $b > a$) is given by

$$A = \int_a^b f(y) dy = \int_a^b x dy$$



2. The area bounded by the straight line $y = a, y = b$ ($a < b$) and the curves $x = f(y)$ and $x = g(y)$, provided $f(y) < g(y)$ (where $a \leq y \leq b$), is given by

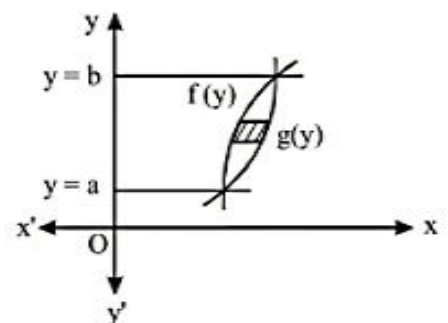
$$A = \int_a^b [g(y) - f(y)] dy$$



3. When two curves $x = f(y)$ and $x = g(y)$ intersect, the bounded area is

$$A = \int_a^b [g(y) - f(y)] dy ; \text{ Where } a < b.$$

where a and b are the roots of the equation $f(y) = g(y)$

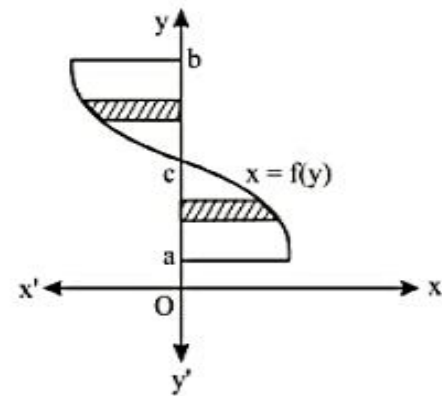


4. If some part of a curve lies left to y -axis, then its area becomes

negative but area cannot be negative. Therefore, we take its modulus.

If the curve crosses the y-axis at c, then the area bounded by the curve $x = f(y)$ and abscissae $y = a$ and $y = b$

$$\begin{aligned} \text{(where } b > a \text{) is given by } A &= \left| \int_a^c f(y) dy \right| + \left| \int_c^b f(y) dy \right| \\ &= A = \int_a^c f(y) dy - \int_c^b f(y) dy \end{aligned}$$



5. The area bounded by $x = f(y)$ and $x = g(y)$ (where $a \leq y \leq b$), when they intersect at $y = c \in (a, b)$ is given by

$$A = \int_a^b |f(y) - g(y)| dy$$

or
$$\int_a^c (f(y) - g(y)) dy + \int_c^b (g(y) - f(y)) dy$$

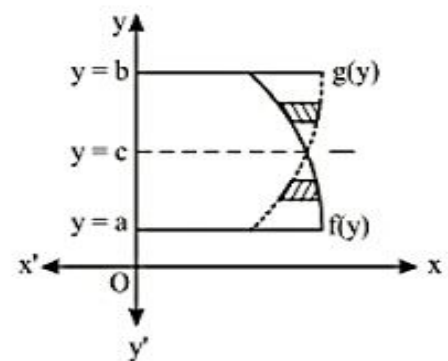


Illustration :

Find the area bounded by the parabola $y = x^2 + 1$ and the straight line $x + y = 3$.

Sol. The two curves meet at points where $3 - x = x^2 + 1$ i.e., $x^2 + x - 2 = 0$
 $\Rightarrow (x + 2)(x - 1) = 0 \Rightarrow x = -2, 1$

$$\therefore \text{required area} = \int_{-2}^1 [(3 - x) - (x^2 + 1)] dx$$

$$= \int_{-2}^1 (2 - x - x^2) dx$$

$$= \left[2x - \frac{x^2}{2} - \frac{x^3}{3} \right]_{-2}^1$$

$$= \left(2 - \frac{1}{2} - \frac{1}{3} \right) - \left(-4 - \frac{4}{2} + \frac{8}{3} \right)$$

$$= \frac{9}{2} \text{ sq. units.}$$

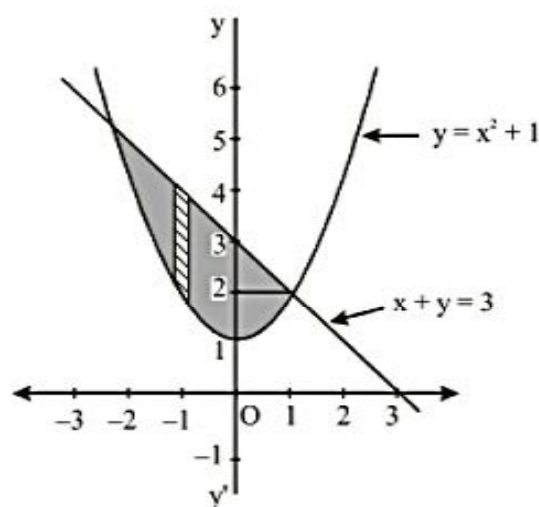


Illustration :

Find the area, lying above x -axis and included between the circle $x^2 + y^2 = 8x$ and the parabola $y^2 = 4x$.

Sol. Solving the curves, we get $x^2 + 4x = 8x \Rightarrow x = 0, 4$.

$$\text{Required area} = \int_0^4 y_{\text{parabola}} dx + \int_4^8 y_{\text{circle}} dx$$

$$\text{Circle is } (x-4)^2 + y^2 = 4^2.$$

$$\text{Area of circle in 1st quadrant} = \frac{1}{4} \pi 4^2 = 4\pi$$

$$A = 2 \int_0^4 \sqrt{x} dx + 4\pi$$

$$= \frac{4}{3} \left[x^{3/2} \right]_0^4 + 4\pi = \frac{2}{3} \times 4\sqrt{4} + 4\pi \text{ sq. units}$$

$$= \frac{32}{3} + 4\pi \text{ sq. units}$$

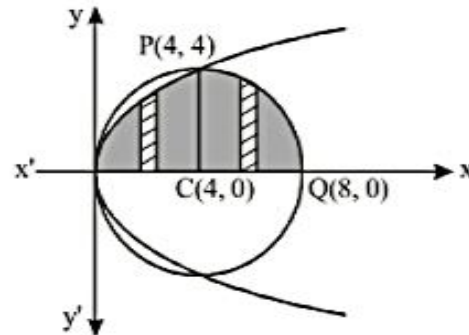


Illustration :

Find the area bounded by the curve $y = (x-1)(x-2)(x-3)$ lying between the ordinates $x = 0$ and $x = 3$.

Sol. $y = (x-1)(x-2)(x-3)$

The curves will intersect the x -axis, when $y = 0$.

$$\Rightarrow (x-1)(x-2)(x-3) = 0$$

$$\Rightarrow x = 1, 2, 3$$

And the curve intersects the y -axis,

$$\text{when } x = 0 \Rightarrow y = -6$$

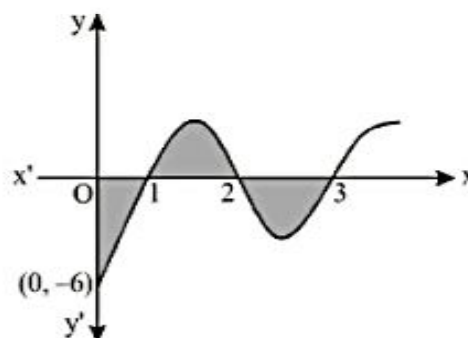
Thus, the graph of the given function for $0 \leq x \leq 3$ is as shown in figure.

Hence, the required area $A =$ shaded area

$$= \left| \int_0^1 y dx \right| + \left| \int_1^2 y dx \right| + \left| \int_2^3 y dx \right|$$

$$\text{Since } \int y dx = \int (x-1)(x-2)(x-3) dx$$

$$= \int (x^3 - 6x^2 + 11x - 6) dx$$



$$= \frac{x^4}{4} - 2x^3 + \frac{11x^2}{2} - 6x$$

$\therefore$ from equation (1)

$$A = \left| \left[\frac{x^4}{4} - 2x^3 + \frac{11x^2}{2} - 6x \right]_0^1 \right| + \left| \left[\frac{x^4}{4} - 2x^3 + \frac{11x^2}{2} - 6x \right]_1^2 \right| + \left| \left[\frac{x^4}{4} - 2x^3 + \frac{11x^2}{2} - 6x \right]_2^3 \right|$$

$$= |-9/4| + (1/4) + |-1/4| = 11/4 \text{ sq. units}$$

Illustration :

Consider the region formed by the lines $x = 0, y = 0, x = 2, y = 2$. Area enclosed by the curves $y = e^x$ and $y = \ln x$, within this region, is being removed. Then, find the area of the remaining region.

Sol. Required area = shaded region

$$= 2 \int_0^{\ln 2} (2 - e^x) dx$$

$$= 2[2x - e^x]_0^{\ln 2}$$

$$= 2(2 \ln 2 - 1) \text{ sq. units}$$

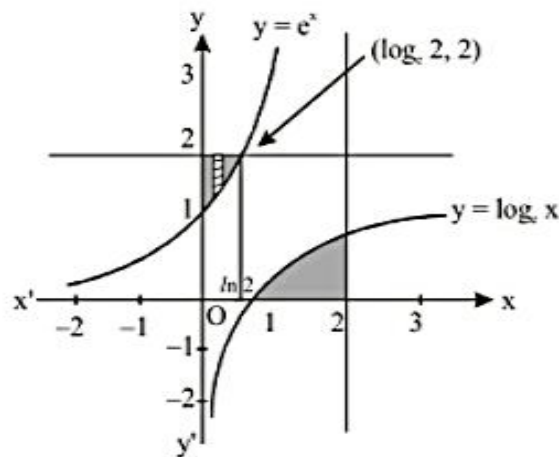


Illustration :

Find the area bounded by the curves $y = \sin x$ and $y = \cos x$ between two consecutive points of their intersection.

Sol. Two consecutive points of intersection of $y = \sin x$ and $y = \cos x$ can be taken as $x = \frac{\pi}{4}$ and $x = \frac{5\pi}{4}$

$$\therefore \text{Required area} = \int_{\pi/4}^{5\pi/4} (\sin x - \cos x) dx$$

$$= [-\cos x - \sin x]_{\pi/4}^{5\pi/4}$$

$$= \frac{2}{\sqrt{2}} + \frac{2}{\sqrt{2}}$$

$$= 2\sqrt{2} \text{ sq. units}$$

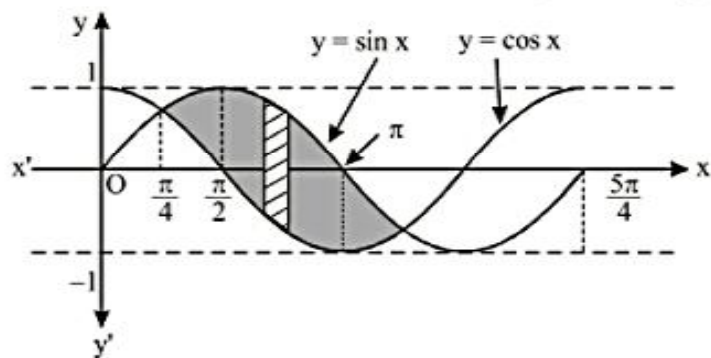


Illustration :

Find the ratio in which the area bounded by the curves $y^2 = 12x$ and $x^2 = 12y$ is divided by the line $x = 3$.

Sol. A_1 = area bounded by $y^2 = 12x$, $x^2 = 12y$ and line $x = 3$

$$\begin{aligned} &= \int_0^3 \sqrt{12x} dx - \int_0^3 \frac{x^2}{12} dx \\ &= \sqrt{12} \left[\frac{2x^{3/2}}{3} \right]_0^3 - \left[\frac{x^3}{36} \right]_0^3 = \frac{45}{4} \text{ sq. units} \end{aligned}$$

$$\begin{aligned} A_2 &= \text{area bounded by } y^2 = 12x \text{ and } x^2 = 12y \\ &= \frac{16(3)(3)}{3} = 48 \text{ sq. units} \end{aligned}$$

$$\therefore \text{required ratio} = \frac{\frac{45}{4}}{48 - \frac{45}{4}} = \frac{45}{147} = \frac{15}{49}$$

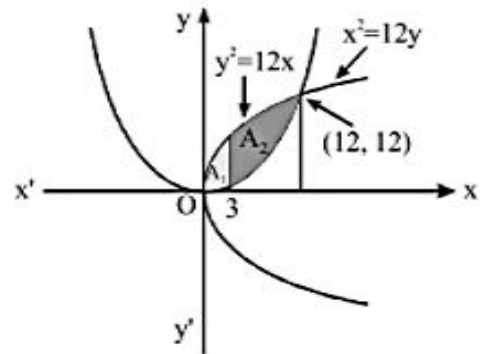


Illustration :

Find the area bounded by

(i) $y = \log_e |x|$ and $y = 0$, (ii) $y = |\log_e |x||$ and $y = 0$

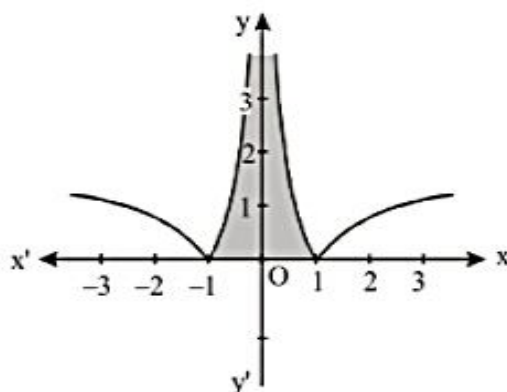
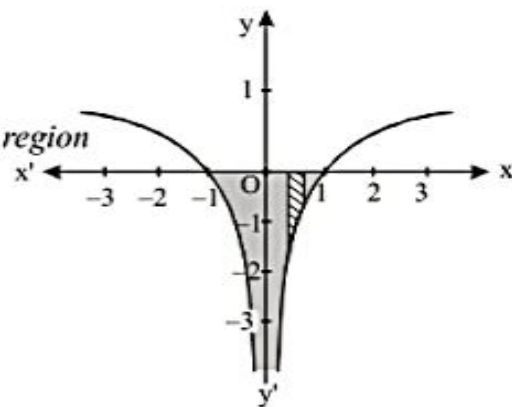
Sol.

(i) $y = \log_e |x|$ and $y = 0$

From the figure, required area = area of the shaded region

$$= 2 \left| \int_0^1 (\log_e x) dx \right| = 2 \left| (x \log_e x - x) \Big|_0^1 \right| = 2 \text{ sq. units}$$

(ii) $y = |\log_e |x||$ and $y = 0$



From the figure, required area = area of the shaded region = $1 + 1 = 2$ sq. units.

Illustration :

Find the area included by the curve $y = \ln x$, x -axis and the two ordinate at $x = \frac{1}{e}$ and $x = e$.

Sol. $A = \left| \int_{1/e}^1 \ln x \, dx \right| + \int_1^e \ln x \, dx = \left| [x(\ln x - 1)]_{1/e}^1 \right| + [x(\ln x - 1)]_1^e = 2 - \frac{2}{e}$

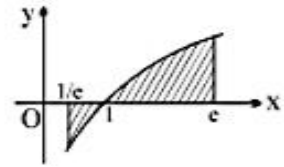


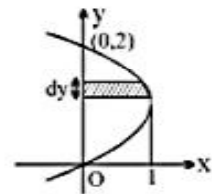
Illustration :

Find the area included by the curve $x = 2y - y^2$ and the y -axis

Sol. Let $x = 2y - y^2$ and the y -axis

$$\frac{dx}{dy} = 2 - 2y = 0 \Rightarrow y = 1 \Rightarrow \text{curve bends at } y = 1;$$

$$A = \int_0^2 x \, dy = \int_0^2 (2y - y^2) \, dy = \left[y^2 - \frac{y^3}{3} \right]_0^2 = 4 - \frac{8}{3} = \frac{4}{3} \text{ Ans.}$$



Alternative method :

This can also be done by taking vertical strip.

$$y^2 - 2y + x = 0$$

$$y = \frac{2 \pm \sqrt{4 - 4x}}{2}$$

$$y = 1 + \sqrt{1 - x} \quad (y_2)$$

$$y = 1 - \sqrt{1 - x} \quad (y_1)$$

$$A = \int_0^1 y \, dx = \int_0^1 2(\sqrt{1 - x}) \, dx$$

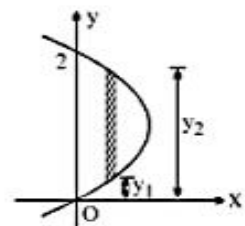


Illustration :

For $b > a > 1$, the area enclosed by the curve $y = \ln x$, y axis and the straight lines $y = \ln a$ and $y = \ln b$ is

(A) $b - a$

(B) $b(\ln b - 1) - a(\ln a - 1)$

(C) $(\ln a)(b - a)$

(D) $(\ln b)(\ln a)$

Sol. Required area = $\int_{\ln a}^{\ln b} e^y dy = [e^y]_{\ln a}^{\ln b} = (b - a)$

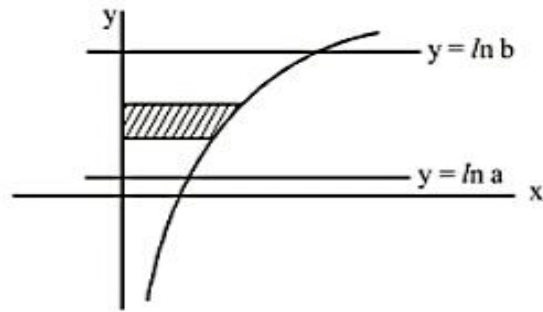


Illustration :

Find the area enclosed between $y = \sin x$; $y = \cos x$ and y -axis in the 1st quadrant

Sol. $A = \int_0^{\pi/4} (\cos x - \sin x) dx = [\sin x + \cos x]_0^{\pi/4} = \sqrt{2} - 1$

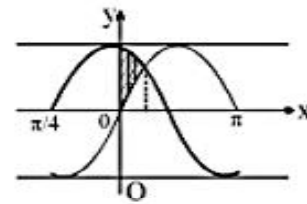
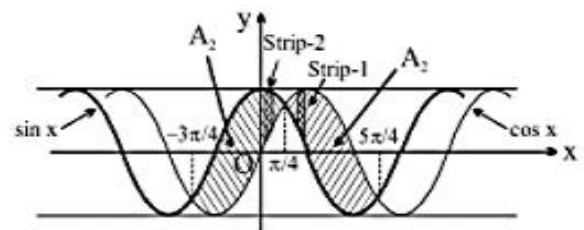


Illustration :

Curves $y = \sin x$; $y = \cos x$ intersect each other at infinite number of points enclosing regions of equal areas. Compute the area of one such equal region.

Sol. (Strip-1) $A_1 = \int_{\pi/4}^{5\pi/4} (\sin x - \cos x) dx = 2\sqrt{2}$

(Strip-2) $A_2 = \int_{-3\pi/4}^{\pi/4} (\cos x - \sin x) dx = 2\sqrt{2}$



So $A_1 = A_2 = 2\sqrt{2}$ sq. unit

Illustration :

Find the area enclosed by $y = \tan x$; $y = \cot x$ and x -axis in 1st quadrant.

Sol. $A = \int_0^{\pi/4} \tan x dx + \int_{\pi/4}^{\pi/2} \cot x dx$

$A = 2 \int_0^{\pi/4} \tan x dx = 2 [\ln |\sec x|]_0^{\pi/4}$

$= 2 \ln \sqrt{2} = \ln 2$

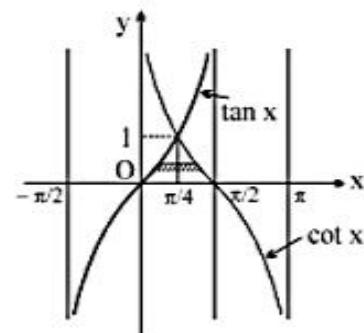


Illustration :

Compute the area enclosed between $y = \tan^{-1}x$; $y = \cot^{-1}x$ and y-axis.

Sol. $A = \int_0^1 (\cot^{-1} x - \tan^{-1} x) dx$

$$A = \int_0^{\frac{\pi}{4}} (\tan y) dy + \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} (\cot y) dy = \ln 2$$

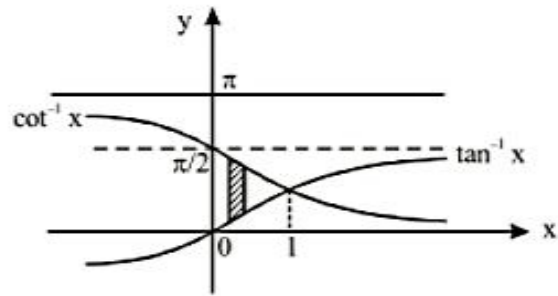


Illustration :

Area enclosed by $y = 9 - x^2$ and coordinates axes.

Sol. $A = \int_0^3 (9 - x^2) dx = 18$

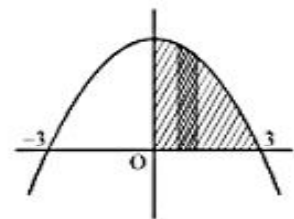


Illustration :

Compute the larger area bounded by $y = 4 + 3x - x^2$ and the coordinates axes.

Sol. $A = \int_0^4 y dx = \int_0^4 (4 + 3x - x^2) dx$

$$= \left[4x + \frac{3}{2}x^2 - \frac{1}{3}x^3 \right]_0^4 = \frac{56}{3}$$

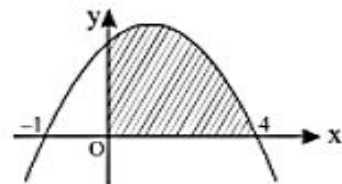


Illustration :

Find the area bounded by $y = \sin^{-1}x$, $y = \cos^{-1}x$ and x-axis.

Sol. $y = \sin^{-1}x$, $y = \cos^{-1}x$ and the x-axis if vertical stripe is used, we get

$$A = \int_0^{1/\sqrt{2}} \sin^{-1} x dx + \int_{1/\sqrt{2}}^1 \cos^{-1} x dx$$

If horizontal strip is used, then

$$A = \int_0^{\pi/4} (\cos y - \sin y) dy$$

$$= [\sin y + \cos y]_0^{\pi/4}$$

$$= \left[\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} - 1 \right] = \sqrt{2} - 1$$

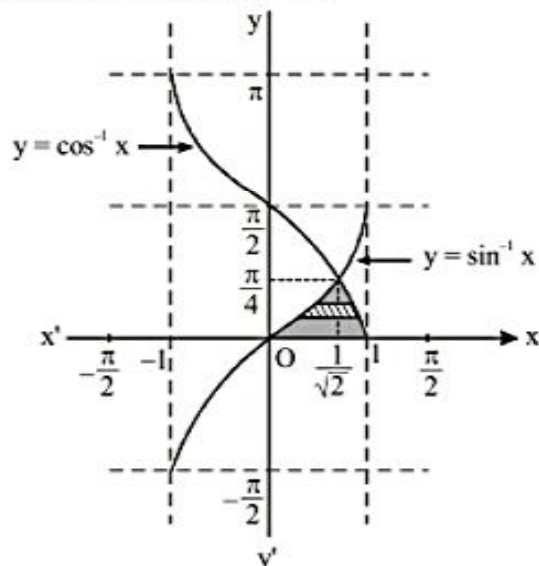


Illustration :

Find the area of the region in the 1st quadrant bounded on the left by the y-axis, below by the line $y = \frac{x}{4}$, above left by the curve $y = 1 + \sqrt{x}$ and above right by the curve $y = \frac{2}{\sqrt{x}}$

Sol. Required area = $\int_0^1 \left(1 + \sqrt{x} - \frac{x}{4}\right) dx + \int_1^4 \left(\frac{2}{\sqrt{x}} - \frac{x}{4}\right) dx$

$$= \left(x + \frac{2}{3}x^{3/2} - \frac{x^2}{8}\right)_0^1 + \left(4\sqrt{x} - \frac{x^2}{8}\right)_1^4$$

$$= \left(1 + \frac{2}{3} - \frac{1}{8}\right) + \left(4 - \frac{15}{8}\right) = \frac{5}{3} + 2 = \frac{11}{3} \text{ sq. units.}$$

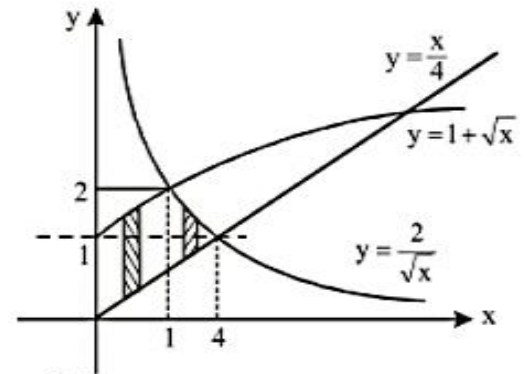


Illustration :

Find the area of the figure bounded by the parabolas $x = -2y^2$, $x = 1 - 3y^2$.

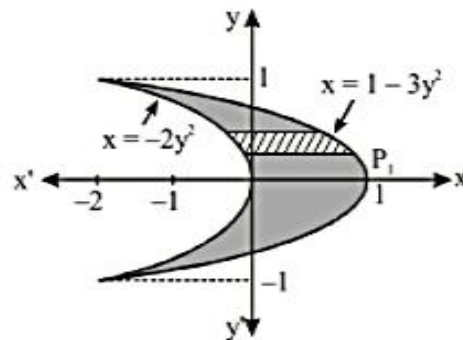
Sol. Solving the equation $x = -2y^2$, $x = 1 - 3y^2$, we find that ordinates of the point of intersection of the two curves as $y_1 = -1$, $y_2 = 1$. The points are $(-2, -1)$ and $(-2, 1)$.

The required area (using horizontal strip)

$$A = 2 \int_0^1 (x_1 - x_2) dy$$

$$= 2 \int_0^1 [(1 - 3y^2) - (-2y^2)] dy$$

$$= 2 \int_0^1 (1 - y^2) dy = 2 \left[y - \frac{y^3}{3} \right]_0^1 = \frac{4}{3}$$



Practice Problem

- Q.1 Find the area lying in the first quadrant and bounded by the curve $y = x^3$ and the line $y = 4x$.
- Q.2 Find the area enclosed by the curves $x^2 = y$, $y = x + 2$ and x-axis.
- Q.3 A curve is given by $y = \begin{cases} \sqrt{4-x^2}, & 0 \leq x < 1 \\ \sqrt{3x}, & 1 \leq x \leq 3 \end{cases}$. Find the area lying between the curve and x-axis.
- Q.4 Find the area of the region bounded by the limits $x = 0$, $x = \frac{\pi}{2}$ and $f(x) = \sin x$, $g(x) = \cos x$.
- Q.5 Find the area bounded by the curve $y = \sin^{-1} x$ and the line $x = 0$, $|y| = \frac{\pi}{2}$.
- Q.6 Find the area bounded by $y = \tan^{-1} x$, $y = \cot^{-1} x$ and y-axis is first quadrant.
- Q.7 Find the area bounded by $y = \log_e x$, $y = -\log_e x$, $y = \log_e(-x)$ and $y = -\log_e(-x)$.
- Q.8 Find the equation of the tangent to the parabola $x^2 = 4y$ with gradient unity. Also find the area enclosed by the curve, the tangent line and
(i) the y-axis (ii) the x-axis
- Q.9 Pair of tangents are drawn from the point $(3, 0)$ on the parabola $y = x^2$. Find the area enclosed by these tangents and the parabola.
- Q.10 Compute the area included between the straight lines, $x - 3y + 5 = 0$; $x + 2y + 5 = 0$ and the circle $x^2 + y^2 = 25$.

Answer key

- | | | |
|-----------------------|---|---|
| Q.1 4 | Q.2 $\frac{5}{6}$ | Q.3 $\frac{1}{6}(2\pi - \sqrt{3} + 36)$ |
| Q.4 $2(\sqrt{2} - 1)$ | Q.5 2 | Q.6 $\log \sqrt{2}$ |
| Q.7 4 | Q.8 $x - y = 1$; P (2, 1); (i) $2/3$; (ii) $1/6$ | |
| Q.9 18 sq. units | Q.10 $\frac{5}{4}(5\pi + 14)$ sq. Units | |

STANDARD AREAS TO BE REMEMBERED :

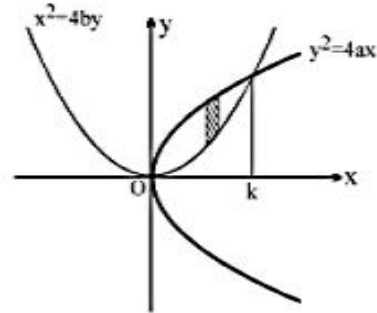
(1) Area bounded by the curve $y^2 = 4ax$; $x^2 = 4by$ is equal to $\frac{16ab}{3}$:

At point of intersection

$$\left(\frac{x^2}{4b}\right)^2 = 4ax \Rightarrow x^4 = 64ab^2x$$

$$\Rightarrow x = 0, (64ab^2)^{1/3}$$

Let $k = 4(ab^2)^{1/3}$



$$A = \int_0^k \left(2\sqrt{a}\sqrt{x} - \frac{x^2}{4b} \right) dx$$

$$= \left[2\sqrt{a} \frac{x^{3/2}}{3/2} - \frac{x^3}{12b} \right]_0^k = \frac{4\sqrt{a}}{3} k^{3/2} - \frac{k^3}{12b} = \frac{4}{3} \sqrt{a} 8(ab^2)^{1/2} - \frac{64(ab^2)}{12b}$$

$$= \frac{32}{3} ab - \frac{16}{3} ab = \frac{16ab}{3}$$

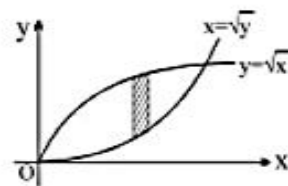
Illustration :

Find the area bounded by the curve $y = \sqrt{x}$; $x = \sqrt{y}$

Sol. $a = \frac{1}{4}$; $b = \frac{1}{4}$

$$\text{Required area} = \frac{16ab}{3} = \frac{16 \cdot \frac{1}{4} \cdot \frac{1}{4}}{3}$$

$$\text{Area} = \frac{1}{3}$$



(2) Area bounded by the parabola $y^2 = 4ax$ and $y = mx$ is equal to $\frac{8a^2}{3m^3}$:

$$y^2 = 4ax \text{ and } y = mx$$

At point of intersection

$$m^2x^2 = 4ax \Rightarrow x = 0, \frac{4a}{m^2}$$

$$\text{Area} = \int_0^c (2\sqrt{a}\sqrt{x} - mx) dx \quad \text{where } c = \frac{4a}{m^2}$$

$$= \left(2\sqrt{a} \frac{x^{\frac{3}{2}}}{\frac{3}{2}} - \frac{mx^2}{2} \right)_0^c = \frac{4\sqrt{a}}{3} c^{\frac{3}{2}} - \frac{mc^2}{2}$$

$$= \frac{4\sqrt{a}}{3} \cdot \frac{8a\sqrt{a}}{m^3} - \frac{m}{2} \cdot \frac{16a^2}{m^4} = \frac{32a^2}{3m^3} - \frac{8a^2}{m^3} = \frac{8a^2}{3m^3}$$

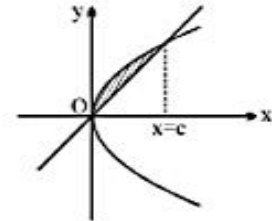


Illustration :

Find the area bounded by the curves $x^2 = y$; $y = |x|$.

$$\text{Sol. Area} = 2 \left(\frac{8a^2}{3m^3} \right) = 16 \left(\frac{\left(\frac{1}{4}\right)^2}{3(1)^3} \right) = \frac{1}{3}$$

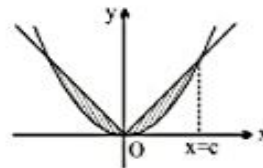
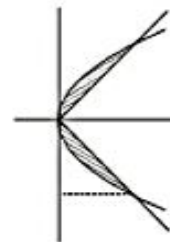


Illustration :

Find the curve bounded $y^2 = x$; $x = |y|$.

$$\text{Sol. Area} = 2 \left(\frac{8a^2}{3m^3} \right) = \frac{16}{3} \cdot \frac{\left(\frac{1}{4}\right)^2}{(1)^3} = \frac{1}{3}$$

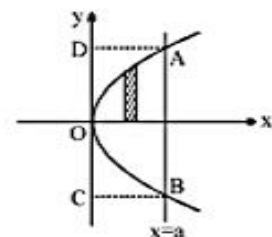


(3) Area enclosed by $y^2 = 4ax$ and its double ordinate at $x = a$:

(chord perpendicular to the axis of symmetry)

Required area = OABO

$$\begin{aligned} &= 2 \cdot \int_0^a (2\sqrt{ax}) dx = 4\sqrt{a} \left(\frac{x^{\frac{3}{2}}}{\frac{3}{2}} \right)_0^a \\ &= \frac{8}{3} \sqrt{a} \cdot (a\sqrt{a}) = \frac{8a^2}{3} \end{aligned}$$



Area of rectangle ABCD = $4a^2$

$$\Rightarrow \boxed{\text{Area of } \triangle AOB = \frac{2}{3} (\text{area } \square ABCD)}$$

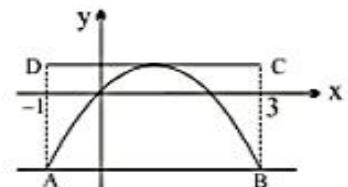
Illustration :

Find the area bounded by the curve. $y = 2x - x^2$, $y + 3 = 0$

Sol. For point of intersection of $y = 2x - x^2$ and $y + 3 = 0$

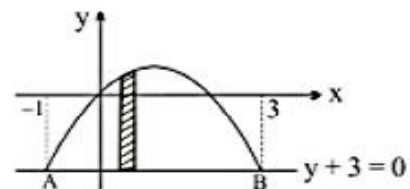
$$\text{Area (ABCD)} = 4 \times 4 = 16$$

$$\text{Required area} = \frac{2}{3} \times 16 = \frac{32}{3}$$



Alternative method :

$$\text{By integration } A = \int_{-1}^3 [(2x - x^2) - (-3)] dx = \frac{32}{3}$$



(4) Whole area of ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is equal to πab :

$$A = 4 \int_0^a \left(b \sqrt{1 - \frac{x^2}{a^2}} \right) dx$$

Put $x = a \sin \theta$

$$A = 4 \int_0^{\pi/2} ab \cos^2 \theta d\theta = 4ab \int_0^{\pi/2} \cos^2 \theta d\theta$$

$$= 4ab \int_0^{\pi/2} \left(\frac{1 + \cos 2\theta}{2} \right) d\theta = 4ab \left(\frac{\pi}{4} \right) = \pi ab$$

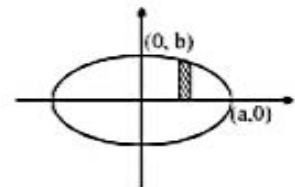


Illustration :

Find the area of ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$.

Sol. Area of ellipse = $\pi ab = \pi (4) (3) = 12\pi$

SHIFTING OF ORIGIN :

Since area remains invariant even if the coordinates axes are shifted, hence shifting of origin in many cases proves to be very convenient in computing the areas.

Illustration :

Area enclosed between the parabolas $y^2 - 2y + 4x + 5 = 0$ and $x^2 + 2x - y + 2 = 0$.

Sol. $y^2 - 2y + 1 \Rightarrow (y - 1)^2 = -4(x + 1)$... (1)

$x^2 + 2x + 1 = y - 1 \Rightarrow (x + 1)^2 = (y - 1)$... (2)

Let $y - 1 = Y$ and $x + 1 = X$

So equation $Y^2 = -4X$ and $X^2 = Y$

$$a = 1, b = \frac{1}{4}$$

$$\text{so required area} = \frac{16ab}{3} = \frac{16}{3} \cdot 1 \cdot \frac{1}{4} = \frac{4}{3}$$

Illustration :

Area enclosed between the ellipse $9x^2 + 4y^2 - 36x + 8y + 4 = 0$ and the line $3x + 2y - 10 = 0$ in the first quadrant.

Sol. $9x^2 + 4y^2 - 36x + 8y + 4 = 0$

$$\Rightarrow 9(x - 2)^2 + 4(y + 1)^2 = 36$$

$$\Rightarrow \frac{(x - 2)^2}{2^2} + \frac{(y + 1)^2}{3^2} = 1 \quad \dots (1)$$

Let $X = x - 2$ and $Y = y + 1$

So equation of ellipse will be

$$\frac{X^2}{2^2} + \frac{Y^2}{3^2} = 1$$

and equation of line $3x + 2y - 10 = 0$... (2)

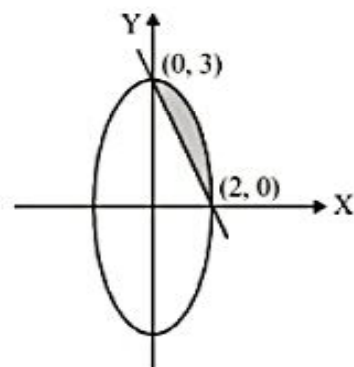
$$3(X + 2) + 2(Y - 1) - 10 = 0$$

$$3X + 2Y - 6 = 0$$

So required area (shaded region)

$$= \frac{\pi ab}{4} - \frac{1}{2}(ab)$$

$$= \frac{\pi}{4}(2)(3) - \frac{1}{2}(2)(3) = \frac{3\pi}{2} - 3 = \frac{3(\pi - 2)}{2}$$



CURVE TRACING :

The approximate shape of a curve, the following procedure in order

(I) SYMMETRY :

(a) Symmetry about x-axis

If the equation of the curve remain unchanged by replacing y by $-y$ then the curve is symmetrical about the x-axis.

e.g., $y^2 = 4ax$.

(b) Symmetry about y-axis

If the equation of the curve remain unchanged by replacing x by $-x$ then the curve is symmetrical about the y-axis.

e.g., $x^2 = 4ay$

(c) Symmetry about both axes

If the equation of the curve remain unchanged by replacing x by $-x$ and y by $-y$ then the curve is symmetrical about the axis of 'x' as well as 'y'.

e.g., $x^2 + y^2 = a^2$

(d) Symmetry about the line $y = x$

If the equation of curve remains unchanged on interchanging 'x' and 'y', then the curve is symmetrical about the line $y = x$

e.g., $x^3 + y^3 = 3xy$.

(II) Find the points where the curve crosses the x-axis and the y-axis.

(III) Find $\frac{dy}{dx}$ and examine, if possible, the intervals where $f(x)$ is increasing or decreasing and also its stationary points.

(IV) Examine y when $x \rightarrow \infty$ or $x \rightarrow -\infty$.

Illustration :

Draw a rough sketch of the curve, $y = \frac{x^2 + 3x + 2}{x^2 - 3x + 2}$ and find the area of the bounded region between the curve and x-axis.

Sol. $f(x) = \frac{(x+1)(x+2)}{(x-1)(x-2)}$

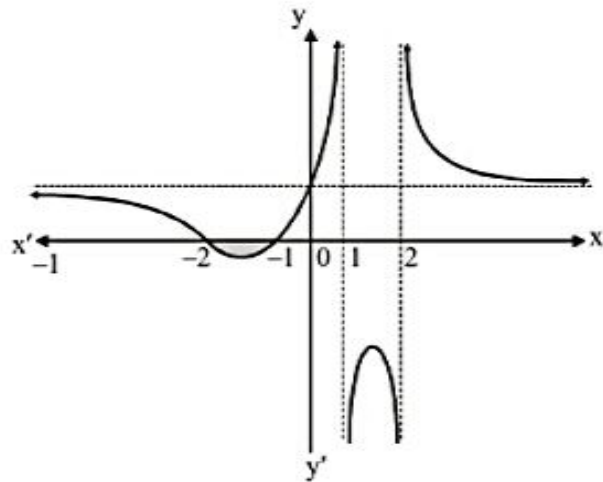
Graph will cut x-axis $x = -1$ and $x = -2$.
It is discontinuous at $x = 1$ and $x = 2$.

$$\lim_{x \rightarrow \pm\infty} f(x) \rightarrow 1, \quad \lim_{x \rightarrow 1^-} f(x) \rightarrow +\infty,$$

$$\lim_{x \rightarrow 1^+} f(x) \rightarrow -\infty,$$

$$\lim_{x \rightarrow 2^-} f(x) \rightarrow -\infty,$$

$$\lim_{x \rightarrow 2^+} f(x) \rightarrow +\infty, f(0) = 1.$$



Now we have to find area of the shaded region. The required area

$$= \left| \int_{-2}^{-1} f(x) dx \right| = \left| \int_{-2}^{-1} \frac{x^2 + 3x + 2}{x^2 - 3x + 2} dx \right| = \left| \int_{-2}^{-1} \left(1 + \frac{6x}{(x-1)(x-2)} \right) dx \right|$$

$$= \left| \left[x \right]_{-2}^{-1} + 6 \int_{-2}^{-1} \left(\frac{2}{x-2} - \frac{1}{x-1} \right) dx \right|$$

$$= \left| 1 + 6 \left[2 \ln |x-2| - \ln |x-1| \right]_{-2}^{-1} \right|$$

$$= \left| 1 + 6 \left[2(\ln 3 - \ln 4) - (\ln 2 - \ln 3) \right] \right| = \left| 1 + 6 \left[3 \ln 3 - 5 \ln 2 \right] \right|$$

$$= 6 \ln \left(\frac{32}{27} \right) - 1 \text{ sq. units}$$

Illustration :

Find the area bounded by the curves $y = -x^2 + 6x - 5$, $y = -x^2 + 4x - 3$ and the straight line $y = 3x - 15$.

Sol. The given curves are

$$y = -x^2 + 6x - 5 \text{ or } (x-3)^2 = -(y-4) \quad \dots(i)$$

which is a parabola with vertex at $A_1(3, 4)$ and axis parallel to negative y-axis. It intersects the x-axis at the point $P(1, 0)$ and $Q(5, 0)$

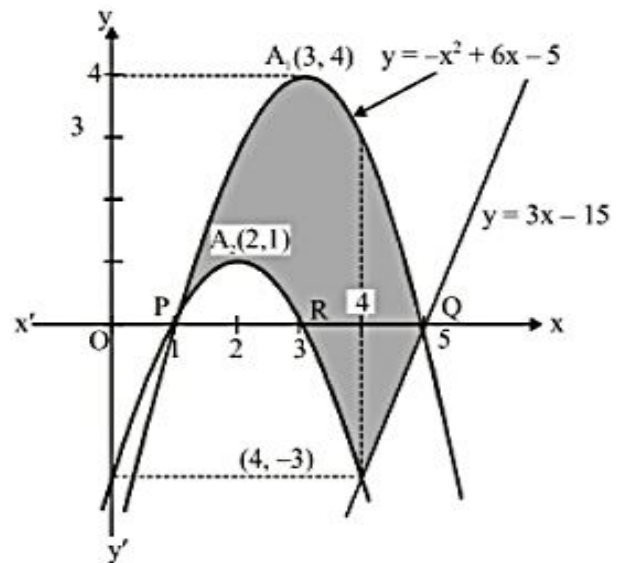
$$y = -x^2 + 4x - 3 \text{ or } (x-2)^2 = -(y-1) \quad \dots(ii)$$

which is parabola with vertex at $A_2(2, 1)$ and axis parallel to negative y -axis. It intersects the x -axis at the points $P(1, 0)$ and $R(3, 0)$.

$$\text{And } y = 3x - 15. \quad \dots(iii)$$

Solving, the points of intersection of (i), (ii) is $(1, 0)$; (i), (iii) are $(-2, -21)$ and $(5, 0)$ and (ii), (iii) are $(-3, -24)$ and $(4, -3)$.

Thus, the required area is the shaded area in the diagram.



$$\begin{aligned} \text{Required area} &= \left| \int_1^4 (y_1 - y_2) dx \right| + \left| \int_4^5 (y_1 - y_3) dx \right| \\ &= \left| \int_1^4 [(-x^2 + 6x - 5) - (-x^2 + 4x - 3)] dx \right| + \left| \int_4^5 [(-x^2 + 6x - 5) - (3x - 15)] dx \right| \\ &= \left| \int_1^4 (2x - 2) dx \right| + \left| \int_4^5 (-x^2 + 3x + 10) dx \right| \\ &= 9 + 19/6 = 73/6 \text{ sq. units.} \end{aligned}$$

Illustration :

The area of the region enclosed by the curves $y = x \log x$ and $y = 2x - 2x^2$ is

- (A) $\frac{7}{12}$ sq. units (B) $\frac{1}{2}$ sq. units (C) $\frac{5}{12}$ sq. units (D) None of these

Sol. Curve tracing : $y = x \log_e x$

Clearly, $x > 0$,

For $0 < x < 1$, $x \log_e x < 0$,

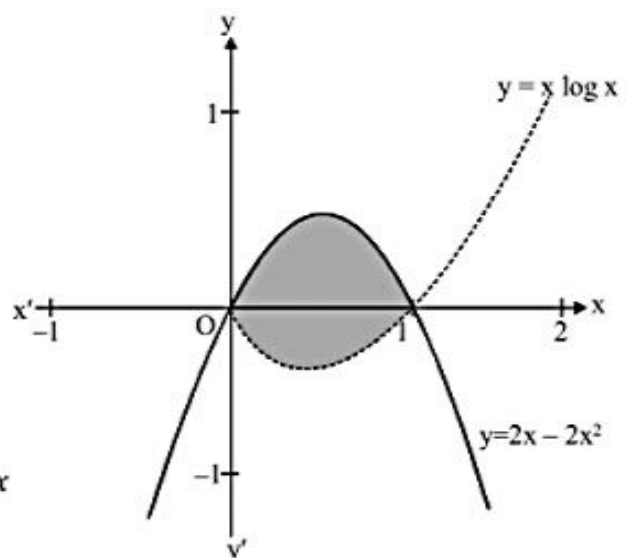
and for $x > 1$, $x \log_e x > 0$

Also $x \log_e x = 0 \Rightarrow x = 1$

$$\text{Further, } \frac{dy}{dx} = 0 \Rightarrow 1 + \log_e x = 0$$

$$\Rightarrow x = 1/e, \text{ which is a point of minima.}$$

$$\text{Required area} = \int_0^1 (2x - 2x^2) dx - \int_0^1 x \log x dx$$



$$= \left[x^2 - \frac{2x^3}{3} \right]_0^1 - \left[\frac{x^2}{2} \log x - \frac{x^2}{4} \right]_0^1$$

$$= \left(1 - \frac{2}{3} \right) - \left[0 - \frac{1}{4} - \frac{1}{2} \lim_{x \rightarrow 0} x^2 \log x \right] = \frac{1}{3} + \frac{1}{4} = \frac{7}{12}$$

Illustration :

Area bounded by $y = \frac{1}{x^2 - 2x + 2}$ and x-axis is

- (A) 2π sq. units (B) $\frac{\pi}{2}$ sq. units (C) 2 sq. units (D) π sq. units

Sol. $y = \frac{1}{(x-1)^2 + 1}$

$$\text{Area} = 2 \int_1^{\infty} \frac{1}{(x-1)^2 + 1} dx$$

$$= 2 \left[\tan^{-1}(x-1) \right]_1^{\infty} = \pi \text{ sq. units}$$

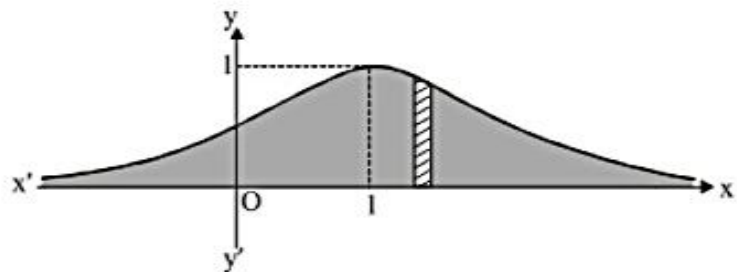


Illustration :

Area bounded by the curve $xy^2 = a^2(a-x)$ and y-axis is

- (A) $\frac{\pi a^2}{2}$ sq. units (B) πa^2 sq. units (C) $3\pi a^2$ sq. units (D) None of these

Sol. $xy^2 = a^2(a-x)$

$$\Rightarrow x = \frac{a^3}{y^2 + a^2}$$

The given curve is symmetrical about x-axis, and meets it at $(a, 0)$.
The line $x = 0$, i.e., y-axis is an asymptote.

$$\text{Area} = 2 \int_0^{\infty} x dy = 2 \int_0^{\infty} \frac{a^3}{y^2 + a^2} dy$$

$$= 2a^3 \frac{1}{a} \left[\tan^{-1} \frac{y}{a} \right]_0^{\infty}$$

$$= 2a^2 \frac{\pi}{2} = \pi a^2 \text{ sq. units.}$$

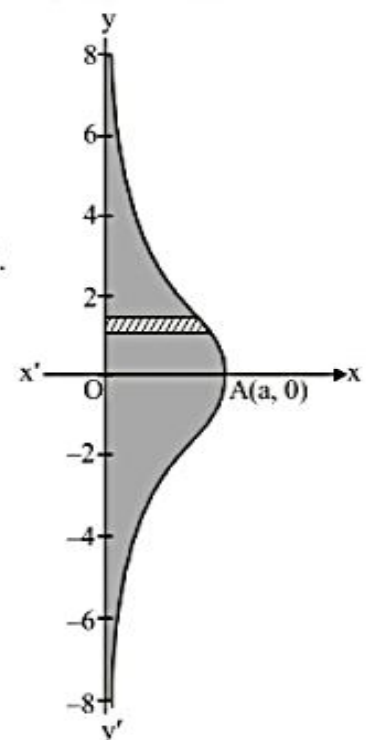


Illustration :

The area between the curve $y = 2x^4 - x^2$, the x-axis and the ordinates of the two minima of the curve is

- (A) $\frac{11}{60}$ sq. units (B) $\frac{7}{120}$ sq. units (C) $\frac{1}{30}$ sq. units (D) $\frac{7}{90}$ sq. units

Sol. The curve is $y = 2x^4 - x^2 = x^2(2x^2 - 1)$
The curve is symmetrical about of axis of y.
The curve passes through the origin and the tangent at the origin is $y = 0$, i.e., x-axis.
The turning points of the curve are given by

$$\frac{dy}{dx} = 8x^3 - 2x = 0 \Rightarrow 2x(4x^2 - 1) = 0$$

$$\Rightarrow x = 0, \pm 1/2$$

$$\text{Now, } \frac{d^2y}{dx^2} = 24x^2 - 2$$

Obviously, $\frac{d^2y}{dx^2}$ is +ve when $x = \pm \frac{1}{2}$ and -ve when $x = 0$

$$x = -1/2 \text{ and } x = 1/2$$

$$\text{At } x = -1/2, \text{ min } y = -1/8.$$

The curve intersects the axes at $O(0, 0)$, $A(-1/\sqrt{2}, 0)$ and $B(1/\sqrt{2}, 0)$.

Thus, the graph of the curve is known in the figure

Here, $y \leq 0$, as x varies from $x = -1/2$ to $x = 1/2$

$\therefore$ The required area = 2 Area OCDO

$$= 2 \left| \int_0^{1/2} y dx \right| = 2 \left| \int_0^{1/2} (2x^4 - x^2) dx \right| = \frac{7}{120} \text{ sq. units.}$$

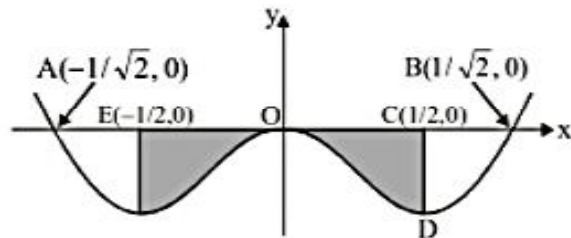


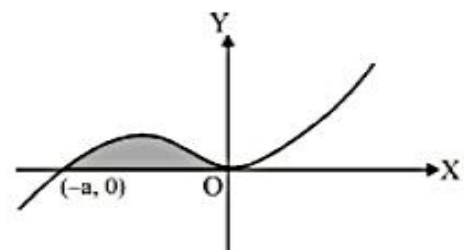
Illustration :

The area bounded by the curve $a^2y = x^2(x + a)$ and x-axis is

- (A) $\frac{a^2}{3}$ sq. units (B) $\frac{a^2}{4}$ sq. units (C) $\frac{3a^2}{4}$ sq. units (D) $\frac{a^2}{12}$ sq. units

Sol. The curve is $y = \frac{x^2(x+a)}{a^2}$, which is a cubic polynomial.

Since $\frac{x^2(x+a)}{a^2} = 0$ has repeated root $x = 0$,



it touches x -axis at $(0, 0)$ and intersects at $(-a, 0)$.

$$\text{Required area} = \int_{-a}^0 y dx = \int_{-a}^0 \left[\frac{x^2(x+a)}{a^2} \right] dx = \frac{a^2}{12} \text{ sq. units.}$$

Illustration :

The area of the loop of the curve, $ay^2 = x^2(a-x)$ is

- (A) $4a^2$ sq. units (B) $\frac{8a^2}{15}$ sq. units (C) $\frac{16a^2}{9}$ sq. units (D) None of these

Sol. $ay^2 = x^2(a-x) \Rightarrow y = \pm x \sqrt{\frac{a-x}{a}}$

Curve tracing : $y = x \sqrt{\frac{a-x}{a}}$

We must have $x \leq a$

For $0 < x \leq a$, $y > 0$ and for $x < 0$, $y < 0$

Also $y = 0 \Rightarrow x = 0, a$

Curve is symmetrical about x -axis.

When $x \rightarrow -\infty$, $y \rightarrow -\infty$

Also, it can be verified that y has only one point of maxima for $0 < x < a$.

$$\text{Area} = 2 \int_0^a x \sqrt{\frac{a-x}{a}} dx$$

$$\sqrt{\frac{a-x}{a}} = t \Rightarrow 1 - \frac{x}{a} = t^2 \Rightarrow x = a(1-t^2)$$

$$\Rightarrow A = 2 \int_1^0 a(1-t^2)t(-2at) dt$$

$$= 4a^2 \int_0^1 (t^2 - t^4) dt = 4a^2 \left[\frac{t^3}{3} - \frac{t^5}{5} \right]_0^1 = 4a^2 \left[\frac{1}{3} - \frac{1}{5} \right] = \frac{8a^2}{15} \text{ sq. units.}$$

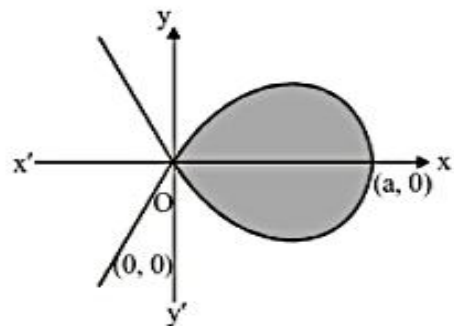


Illustration :

The area of the region enclosed between the curves $x = y^2 - 1$ and $x = |y| \sqrt{1-y^2}$ is

- (A) 1 sq. units (B) $\frac{4}{3}$ sq. units (C) $\frac{2}{3}$ sq. units (D) 2 sq. units

Sol. $A = 2 \int_0^1 [y\sqrt{1-y^2} - (y^2 - 1)] dy$
 $= 2 \text{ sq. units}$

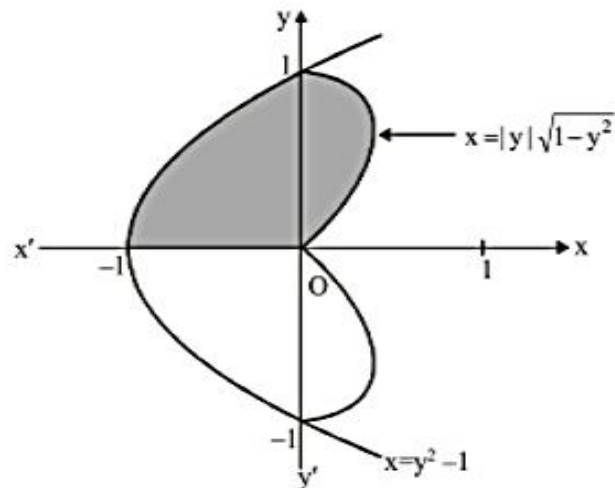


Illustration :

The area bounded by the loop of the curve $4y^2 = x^2(4 - x^2)$ is

- (A) $\frac{7}{3}$ sq. units (B) $\frac{8}{3}$ sq. units (C) $\frac{11}{3}$ sq. units (D) $\frac{16}{3}$ sq. units

Sol. $4y^2 = x^2(4 - x^2)$

$$\Rightarrow y = \pm \frac{1}{2} \sqrt{x^2(4 - x^2)}$$

$$\Rightarrow y = \pm \frac{x}{2} \sqrt{(4 - x^2)}$$

$$\therefore \text{Area (A)} = 4 \int_0^2 \frac{x}{2} \sqrt{(4 - x^2)} dx$$

Let $4 - x^2 = t \Rightarrow -2x dx = dt$

$$\Rightarrow A = \frac{-4}{4} \int_4^0 \sqrt{t} dt = \int_0^4 \sqrt{t} dt = \left[\frac{t^{3/2}}{3/2} \right]_0^4 = \frac{2}{3} \times [\sqrt{64} - 0]$$

$$\Rightarrow A = \frac{16}{3} \text{ sq. units.}$$

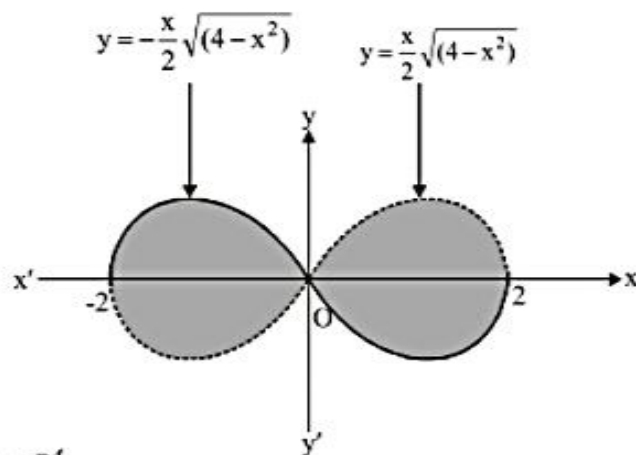


Illustration :

The area enclosed by the curves, $xy^2 = a^2(a - x)$ and $(a - x)y^2 = a^2x$ is

- (A) $(\pi - 2)a^2$ sq. units (B) $(4 - \pi)a^2$ sq. units
 (C) $\pi \frac{a^2}{3}$ sq. units (D) None of these

Sol. The two curves are

$$xy^2 = a^2(a-x) \Rightarrow x = \frac{a^3}{a^2+y^2} \quad \dots(1)$$

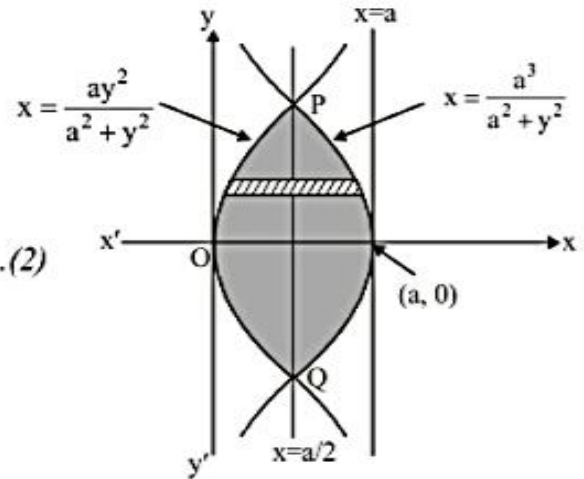
$$\text{and } (a-x)y^2 = a^2x$$

$$\Rightarrow x = \frac{ay^2}{a^2+y^2} = \frac{ay^2+a^3-a^3}{a^2+y^2} = a - \frac{a^3}{a^2+y^2} \quad \dots(2)$$

Curve (1) is symmetrical about x-axis, and have y-axis as the asymptote.

Curve (2) is symmetrical about x-axis, tangent at origin as y-axis and the asymptote $x = a$.

The two curves intersect at the point $P(a/2, a)$ and $Q(a/2, -a)$.



$$\begin{aligned} \text{Required area} &= 2 \int_0^a \left(\frac{a^3}{a^2+y^2} - \frac{ay^2}{a^2+y^2} \right) dy = 2a \int_0^a \frac{a^2-y^2}{a^2+y^2} dy = 2a \left[2 \int_0^a \frac{a^2}{a^2+y^2} dy - \int_0^a dy \right] \\ &= 2a \left[2a \tan^{-1} \left(\frac{y}{a} \right) \Big|_0^a - a \right] = 2a \left[2a \frac{\pi}{4} - a \right] = a^2 (\pi - 2) \end{aligned}$$

Illustration :

The area bounded by the curves $y = xe^x$, $y = xe^{-x}$ and the line $x = 1$ is

- (A) $\frac{2}{e}$ sq. units (B) $1 - \frac{2}{e}$ sq. units (C) $\frac{1}{e}$ sq. units (D) $1 - \frac{1}{e}$ sq. units

Sol. Curve tracing : $y = xe^x$

$$\text{Let } \frac{dy}{dx} = 0 \Rightarrow e^x + xe^x = 0 \Rightarrow x = -1.$$

Also, at $x = -1$,

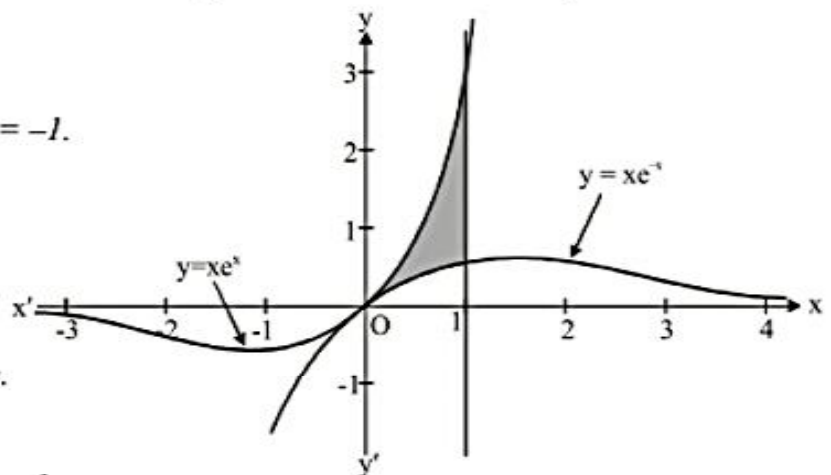
$\frac{dy}{dx}$ changes sign from -ve to +ve

hence, $x = -1$ is a point of minima.

When $x \rightarrow \infty$, $y \rightarrow \infty$

$$\text{Also } \lim_{x \rightarrow -\infty} xe^x = \lim_{x \rightarrow -\infty} \frac{x}{e^{-x}} = \lim_{x \rightarrow -\infty} \frac{1}{e^{-x}} = 0$$

With similar types of arguments, we can draw the graph of $y = xe^{-x}$.



$$\begin{aligned}\text{Required area} &= \int_0^1 x e^x dx - \int_0^1 x e^{-x} dx = [x e^x]_0^1 - \int_0^1 e^x dx - \left([-x e^{-x}]_0^1 + \int_0^1 e^{-x} dx \right) \\ &= e - (e - 1) - \left(-e^{-1} - (e^{-1} - 1) \right) = \frac{2}{e} \text{ sq. units}\end{aligned}$$

Illustration :

The area bounded by the two branches of curve $(y - x)^2 = x^3$ and the straight line $x = 1$ is

- (A) $\frac{1}{5}$ sq. units (B) $\frac{3}{5}$ sq. units (C) $\frac{4}{5}$ sq. units (D) $\frac{8}{4}$ sq. units

Sol. $(y - x)^2 = x^3$, where $x \geq 0$

$$\Rightarrow y - x = \pm x^{3/2}$$

$$\Rightarrow y = x + x^{3/2} \quad \dots(1)$$

$$y = x - x^{3/2} \quad \dots(2)$$

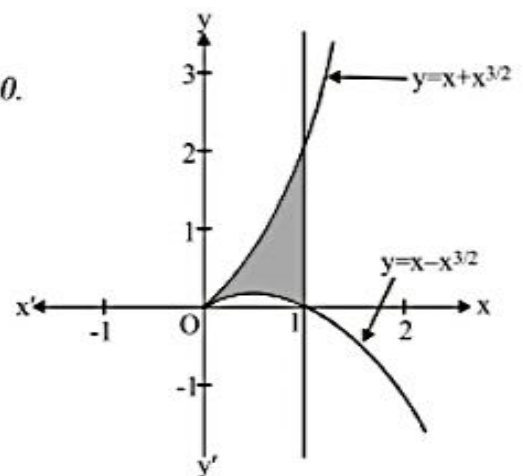
Function (1) is an increasing function.

Function (2) meets x-axis, where $x - x^{3/2} = 0$ or $x = 0, 1$.

Also, for $0 < x < 1$, $x - x^{3/2} > 0$ and for $x > 1$, $x - x^{3/2} < 0$.

When $x \rightarrow \infty$, $y \rightarrow -\infty$,

From these information, we can plot the graph as below:



$$\text{Required area} = \int_0^1 [(x + x^{3/2}) - (x - x^{3/2})] dx$$

$$= 2 \int_0^1 x^{3/2} dx = 2 \left[\frac{x^{5/2}}{5/2} \right]_0^1 = \frac{4}{5} \text{ sq. units}$$

Illustration :

- (a) Sketch and find the area bounded by the curve $\sqrt{|x|} + \sqrt{|y|} = \sqrt{a}$ and $x^2 + y^2 = a^2$ (where $a > 0$).
 (b) If curve $|x| + |y| = a$ divides the area in two parts, then find their ratio in first quadrant only.

Sol. $\sqrt{|x|} + \sqrt{|y|} = \sqrt{a}$

$x = 0 \Rightarrow y = \pm a$

$y = 0 \Rightarrow x = \pm a$

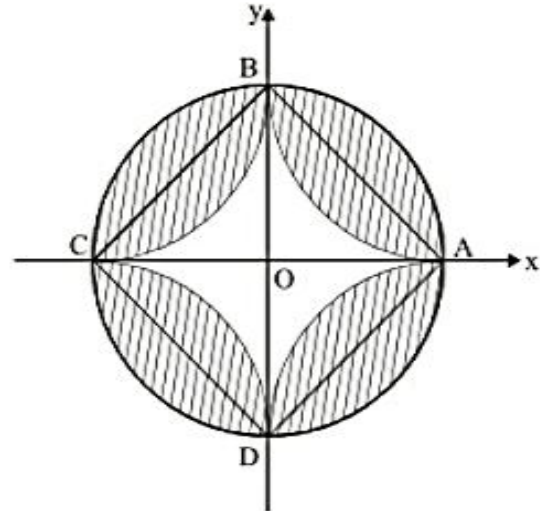
(a) Required area

$$4 \int_0^a \sqrt{a^2 - x^2} dx - 4 \int_0^a (\sqrt{a} - \sqrt{x})^2 dx$$

$$= \pi a^2 - 4 \int_0^a (\sqrt{a} - \sqrt{x})^2 dx$$

$$= \pi a^2 - 4 \int_0^a [a + x - 2\sqrt{a}\sqrt{x}] dx = \pi a^2 - 4 \left[a^2 + \frac{a^2}{2} - 2\sqrt{a} \frac{2}{3} a^{3/2} \right]$$

$$= \pi a^2 - \left[\frac{3a^2}{2} - \frac{4}{3} a^2 \right] = \pi a^2 - 4 \frac{a^2}{6} = \left(\pi - \frac{2}{3} \right) a^2 \text{ sq. units.}$$



(b) Area included between curves and circle in 1st quadrant $= \frac{1}{4} \pi a^2 - \frac{1}{2} a^2 = \frac{(\pi - 2)a^2}{4}$

Area included between $|x| + |y| = a$ and curve $\sqrt{|x|} + \sqrt{|y|} = \sqrt{a}$ in 1st quadrant

$$\frac{1}{4} \left(\pi - \frac{2}{3} \right) a^2 - \left(\frac{\pi}{4} - \frac{1}{2} \right) a^2 = \frac{a^2}{3}$$

$$\text{Area ratio} = \frac{4}{3(\pi - 2)}$$

Illustration :

Area enclosed between the curves $y = ex \cdot \ln x$ and $y = \frac{\ln x}{ex}$

Sol. $y = ex \ln x$

$$\frac{dy}{dx} = e(1 + \ln x) \quad \therefore \quad \frac{dy}{dx} = 0 \quad \Rightarrow \quad x = e^{-1}$$

$$\frac{d^2y}{dx^2} = \frac{e}{x} \quad \therefore \quad \left. \frac{d^2y}{dx^2} \right|_{x=\frac{1}{e}} > 0$$

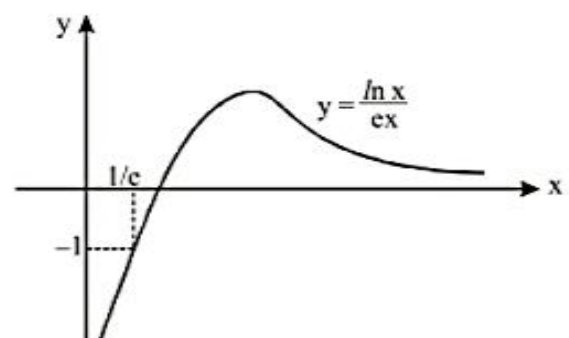
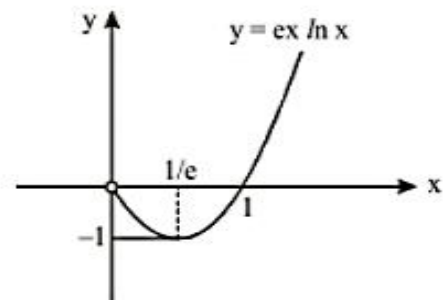
$$\Rightarrow \text{minimum at } x = \frac{1}{e}$$

$$y = \frac{\ln x}{ex}$$

$$\frac{dy}{dx} = \frac{1}{e} \left(\frac{1 - \ln x}{x^2} \right) \quad \therefore \quad \frac{dy}{dx} = 0 \text{ when } x = e$$

$$\left. \frac{dy}{dx} \right|_{x=e^+} < 0, \quad \left. \frac{dy}{dx} \right|_{x=e^-} > 0$$

$$\Rightarrow \text{at } x = e, \quad y = \frac{\ln x}{ex} \text{ has local maxima}$$



$$\text{Required area} = \int_{1/e}^1 \left(\frac{\ln x}{ex} - ex \ln x \right) dx$$

$$= \left[\frac{(\ln x)^2}{2e} - e \left(\frac{x^2}{2} \ln x - \frac{x^2}{4} \right) \right]_{1/e}^1 = \frac{e}{4} - \left[\frac{1}{2e} - e \left(\frac{-1}{2e^2} - \frac{1}{4e^2} \right) \right]$$

$$= \frac{e}{4} - \left[\frac{1}{2e} + \frac{3}{4e} \right] = \frac{e}{4} - \frac{5}{4e} = \left(\frac{e^2 - 20}{4e} \right)$$

Illustration :

Area enclosed by the curve $(y - \sin^{-1} x)^2 = x - x^2$.

Sol. $(y - \sin^{-1} x)^2 = x - x^2$
 $y = \sin^{-1} x \pm \sqrt{x - x^2} \Rightarrow \text{domain } x \in [0, 1]$

Area enclosed by the curve

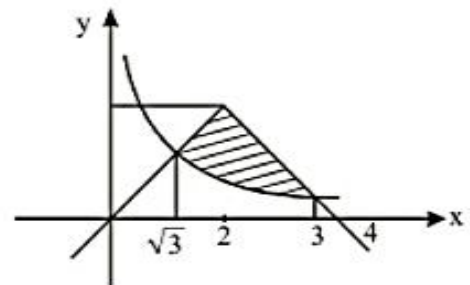
$$\begin{aligned} &= \int_0^1 \left(\sin^{-1} x + \sqrt{x - x^2} \right) - \left(\sin^{-1} x - \sqrt{x - x^2} \right) dx = 2 \int_0^1 \sqrt{x - x^2} dx \\ &= 2 \int_0^1 \sqrt{\frac{1}{4} - \left(x - \frac{1}{2}\right)^2} dx = 2 \left[\frac{1}{2} \left(x - \frac{1}{2}\right) \sqrt{x - x^2} + \frac{1}{2} \left(\frac{1}{4}\right) \sin^{-1} \left(\frac{x - \frac{1}{2}}{\frac{1}{2}}\right) \right]_0^1 \\ &= 2 \left[\left(0 + \frac{1}{8} \frac{\pi}{2}\right) - \left(0 + \frac{1}{8} \left(-\frac{\pi}{2}\right)\right) \right] = 2 \left(\frac{\pi}{16} + \frac{\pi}{16} \right) = \frac{\pi}{4} \end{aligned}$$

Illustration :

Area of the closed figure bounded by the curves $y = 2 - |2 - x|$ and $y = \frac{3}{|x|}$

Sol. $y_1 = 2 - |2 - x| = \begin{cases} x; & x \leq 2 \\ 4 - x; & x > 2 \end{cases}$

and $y_2 = \begin{cases} \frac{3}{x}; & x > 0 \\ -\frac{3}{x}; & x < 0 \end{cases}$



so $\frac{3}{x} = x \Rightarrow x = \sqrt{3}$

and $\frac{3}{x} = 4 - x \Rightarrow 3 = 4x - x^2 \Rightarrow x^2 - 4x + 4 = 1$

$\Rightarrow x - 2 = \pm 1 \Rightarrow x = 3$

so required area = $\int_{\sqrt{3}}^2 \left(x - \frac{3}{x}\right) dx + \int_2^3 \left(4 - x - \frac{3}{x}\right) dx$

$= \left[\frac{x^2}{2} - 3 \ln x \right]_{\sqrt{3}}^2 + \left[4x - \frac{x^2}{2} - 3 \ln x \right]_2^3$

$= 2 - 3 \ln 2 - \frac{3}{2} + 3 \ln(\sqrt{3}) + \left(12 - \frac{9}{2} - 3 \ln 3\right) - \left(8 - \frac{4}{2} - 3 \ln 2\right)$

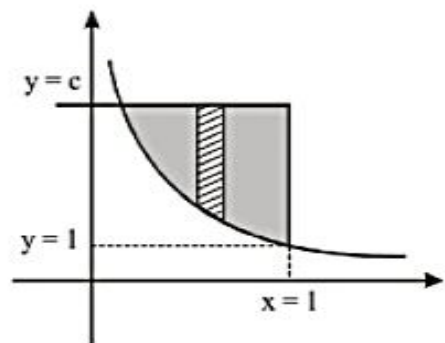
$$= 2 - 3 \ln 2 - \frac{3}{2} + \frac{3}{2} \ln 3 + 4 - \frac{5}{2} - 3 \ln 3 + 3 \ln 2$$

$$= \frac{1}{2} + \frac{3}{2} - \frac{3}{2} \ln 3 = \frac{4 - 3 \ln 3}{2}$$

DETERMINATION OF PARAMETERS :

Illustration :

Find the value of c for which the area of the figure bounded by the curves $y = \frac{4}{x^2}$; $x = 1$ and $y = c$ is equal to $\frac{9}{4}$.



Sol. So required area = $\int_{2/\sqrt{c}}^1 \left(c - \frac{4}{x^2} \right) dx = \left[cx + \frac{4}{x} \right]_{2/\sqrt{c}}^1$

$$\text{Area} = c \left(1 - \frac{2}{\sqrt{c}} \right) + 4 - 2\sqrt{c} = c - 4\sqrt{c} + 4 = \frac{9}{4}$$

$$\Rightarrow (\sqrt{c} - 2)^2 = \frac{9}{4} \Rightarrow \sqrt{c} = 2 \pm \frac{3}{2}$$

$$\sqrt{c} = \frac{1}{2}, \frac{7}{2}$$

$$c = \frac{1}{4}, \frac{49}{4}$$

Illustration :

If the area bounded by $y = x^2 + 2x - 3$ and the line $y = kx + 1$ is the least, find k and also the least area.

Sol. x_1 and x_2 are the roots of the equation

$$x^2 + 2x - 3 = kx + 1, \text{ or}$$

$$x^2 + (2 - k)x - 4 = 0$$

$$\Rightarrow \begin{cases} x_1 + x_2 = k - 2 \\ x_1 x_2 = -4 \end{cases}$$

$$A = \int_{x_1}^{x_2} [kx + 1 - (x^2 + 2x - 3)] dx$$

$$= \left[(k-2) \frac{x^2}{2} - \frac{x^3}{2} + 4x \right]_{x_1}^{x_2} = \left[(k-2) \frac{x_2^2 - x_1^2}{2} - \frac{1}{3} (x_2^3 - x_1^3) + 4(x_2 - x_1) \right]$$

$$\begin{aligned}
 &= (x_2 - x_1) \left[\frac{(k-2)^2}{2} - \frac{1}{3}((x_2 + x_1)^2 - x_1 x_2) + 4 \right] \\
 &= \sqrt{(x_2 + x_1)^2 - 4x_1 x_2} \left[\frac{(k-2)^2}{2} - \frac{1}{3}((k-2)^2 + 4) + 4 \right] \\
 &= \frac{\sqrt{(k-2)^2 + 16}}{6} \left[\frac{1}{6}(k-2)^2 + \frac{8}{3} \right] \\
 &= \frac{[(k-2)^2 + 16]^{3/2}}{6}
 \end{aligned}$$

which is least when $k = 2$ and $A_{\text{least}} = 32/3$ sq. units.

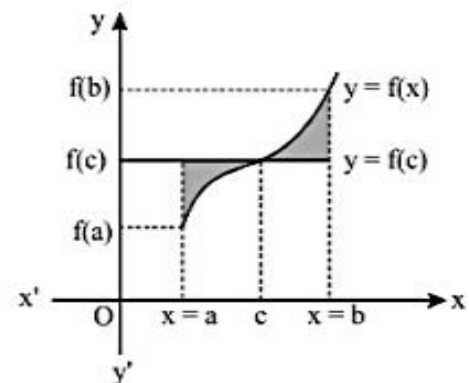
VARIABLE AREA GREATEST AND LEAST VALUE :

An important concept :

If $y = f(x)$ is a monotonic function in (a, b) then the area bounded by the ordinates at $x = a, x = b$,

$y = f(x)$ and $y = f(c)$, [where $c \in (a, b)$] is minimum when $c = \frac{a+b}{2}$.

Proof:
$$\begin{aligned}
 A &= \int_a^c (f(c) - f(x)) dx + \int_c^b (f(x) - f(c)) dx \\
 &= f(c)(c-a) - \int_a^c f(x) dx + \int_c^b f(x) dx - f(c)(b-c) \\
 A &= [2c - (a+b)] f(c) + \int_c^b f(x) dx - \int_a^c f(x) dx
 \end{aligned}$$



Differentiating w.r.t. c ,

$$\frac{dA}{dc} = [2c - (a+b)] f'(c) + 2f(c) + 0 - f(c) - (f(c))$$

for maxima and minima $\frac{dA}{dc} = 0$

$$\Rightarrow f'(x)[2c - (a+b)] = 0 \text{ (as } f'(c) \neq 0 \text{)}$$

hence $c = \frac{a+b}{2}$

Also $c < \frac{a+b}{2}, \frac{dA}{dc} < 0$ and $c > \frac{a+b}{2}, \frac{dA}{dc} > 0$.

Hence A is minimum when $c = \frac{a+b}{2}$.

Note : Let $f(x)$ be the bijective function and $g(x)$ be the inverse of it then area bounded by $y = g(x)$, and the ordinate at $x = a$ and $x = b$ is same as area bounded by $y = f(x)$ and the abscissa at $y = a$ and $y = b$ as $f(x)$ and $g(x)$ are mirror image with respect to line $y = x$.

Illustration :

If the area bounded by $f(x) = \frac{x^3}{3} - x^2 + a$ and the straight lines $x = 0 ; x = 2$ and the x -axis is minimum then find the value of 'a'.

Sol. $f(x) = \frac{x^3}{3} - x^2 + a$

$f'(x) = x^2 - 2x = x(x - 2) < 0$ (note that $f(x)$ is monotonic in $(0, 2)$)

Hence for the minimum and $f(x)$ must cross the x -axis at $\frac{0+2}{2} = 1$

Hence $f(1) = \frac{1}{3} - 1 + a = 0$

$\Rightarrow a = \frac{2}{3}$

Illustration :

The value of the parameter a for which the area of the figure bounded by the abscissa axis, the graph of the function $y = x^3 + 3x^2 + x + a$ and the straight lines, which are parallel to the axis of ordinates and cut the abscissa axis at the point of extremum of the function, is the least, is

- (A) 2 (B) 0 (C) -1 (D) 1

Sol. $f(x) = x^3 + 3x^2 + x + a$
 $f'(x) = 3x^2 + 6x + 1 = 0$

$\Rightarrow x = -1 \pm \frac{\sqrt{6}}{3}$

Hence, $f(x)$ cuts the x -axis at $\frac{1}{2} \left[\left(-1 + \frac{\sqrt{6}}{3} \right) + \left(-1 - \frac{\sqrt{6}}{3} \right) \right] = -1$.

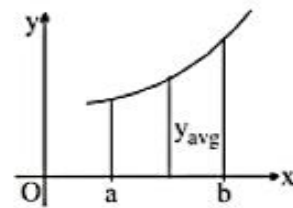
$$f(-1) = -1 + 3 - 1 + a = 0$$

$$a = -1.$$

AVERAGE VALUE OF A FUNCTION :

Average value of the function in $y = f(x)$
w.r.t. x over an interval $a \leq x \leq b$ is defined as

$$y_{\text{avg}} = \frac{1}{b-a} \int_a^b f(x) dx$$



Note :

- (i) Average value can be +ve, -ve or zero.
- (ii) If the function is defined in $(0, \infty)$ then

$$y_{\text{avg}} = \lim_{b \rightarrow \infty} \frac{1}{b} \int_0^b f(x) dx \quad \text{provided the limit exists.}$$

Root mean square value (RMS) is defined as

$$\rho = \left[\frac{1}{b-a} \int_a^b f^2(x) dx \right]^{\frac{1}{2}}$$

Illustration :

Compute the average value of $f(x) = \frac{\cos^2 x}{\sin^2 x + 4 \cos^2 x}$ in $\left[0, \frac{\pi}{2} \right]$

Sol. Average value of $f(x) = \frac{\cos^2 x}{\sin^2 x + 4 \cos^2 x}$

$$y_{\text{average}} = \frac{1}{\left(\frac{\pi}{2} - 0 \right)} \int_0^{\pi/2} \frac{\cos^2 x}{(\sin^2 x + 4 \cos^2 x)} dx = \frac{2}{\pi} \int_0^{\pi/2} \frac{1}{(\tan^2 x + 4)} dx$$

$$= \frac{2}{\pi} \int_0^{\pi/2} \frac{\sec^2 x}{\sec^2 x (\tan^2 x + 4)} dx = \frac{2}{\pi} \int_0^{\pi/2} \frac{\sec^2 x dx}{(1 + \tan^2 x)(4 + \tan^2 x)}$$

Put $t = \tan x$ so $dt = \sec^2 x dx$

$$\begin{aligned} y_{\text{average}} &= \frac{2}{\pi} \int_0^{\infty} \frac{dt}{(t^2 + 1)(4 + t^2)} = \frac{2}{3\pi} \int_0^{\infty} \left[\frac{1}{t^2 + 1} - \frac{1}{t^2 + 4} \right] dt \\ &= \frac{2}{3\pi} \left[\tan^{-1} t - \frac{1}{2} \tan^{-1} \left(\frac{t}{2} \right) \right]_0^{\infty} = \frac{2}{3\pi} \left[\frac{\pi}{2} - \frac{\pi}{4} \right] = \frac{2}{3\pi} \frac{\pi}{4} = \frac{1}{6} \quad \text{Ans.} \end{aligned}$$

DETERMINATION OF FUNCTION :

The area function $A(x)$ satisfies the differential equation $\frac{dA(x)}{dx} = f(x)$ with initial condition $A(a) = 0$
i.e. derivative of the area function is the function itself.

Note :

If $F(x)$ is any integral of $f(x)$ then,

$$A(x) = \int f(x) dx = [F(x) + c] \quad A(a) = 0 = F(a) + c \Rightarrow c = -F(a)$$

hence $A(x) = F(x) - F(a)$. Finally by taking $x = b$ we get, $A(b) = F(b) - F(a)$.

Illustration :

The area from 0 to x under a certain graph is given to be $A = \sqrt{1+3x} - 1, x \geq 0$;

- Find the average rate of change of A w.r.t. x as x increases from 1 to 8.
- Find the instantaneous rate of change of A w.r.t. x at $x = 5$.
- Find the ordinate (height) y of the graph as a function of x .
- Find the average value of the ordinate (height) y , w.r.t. x as x increases from 1 to 8

Sol. $A = \sqrt{1+3x} - 1 = \int_0^x f(x) dx$

$$\begin{aligned} (a) \quad \frac{dA}{dx} \Big|_{\text{avg}} &= \frac{1}{(8-1)} \int_1^8 \left(\frac{dA}{dx} \right) dx \\ &= \frac{1}{7} \left(\sqrt{1+3x} - 1 \right) \Big|_1^8 = \frac{1}{7} (4 - 1) = \frac{3}{7} \end{aligned}$$

$$(b) \quad \left. \frac{dA}{dx} \right|_{x=5} = \frac{3}{2\sqrt{1+3x}} = \frac{3}{2\sqrt{1+3(5)}} = \frac{3}{8}$$

$$(c) \quad A(x) = \int_0^x f(x) dx = \sqrt{1+3x} - 1$$

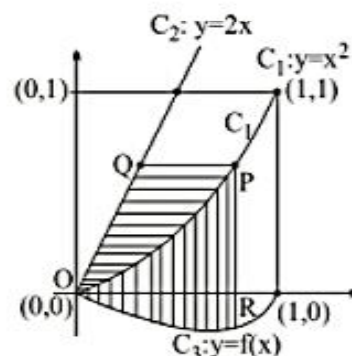
Differentiating w.r.t. x

$$f(x) = \frac{3}{2\sqrt{1+3x}}$$

$$(d) \quad y_{avg} = \frac{1}{(8-1)} \int_1^8 f(x) dx = \frac{1}{7} (\sqrt{1+3x} - 1)_1^8 = \frac{3}{7}$$

Illustration :

Let C_1 & C_2 be the graphs of the functions $y = x^2$ & $y = 2x$, $0 \leq x \leq 1$ respectively. Let C_3 be the graph of a function $y = f(x)$, $0 \leq x \leq 1$, $f(0) = 0$. For a point P on C_1 , let the lines through P , parallel to the axes, meet C_2 & C_3 at Q & R respectively (see figure). If for every position of P (on C_1), the areas of the shaded regions OPQ & ORP are equal, determine the function $f(x)$. [JEE '98, 8]



Sol. Let $P(h, h^2)$ be a point on the curve C_1 .

$$\Rightarrow R(h, f(h))$$

$$\text{Area } OPQO = \text{Area } OPRO$$

$$\int_0^{h^2} \left(\sqrt{y} - \frac{y}{2} \right) dy = \int_0^h (x^2 - f(x)) dx$$

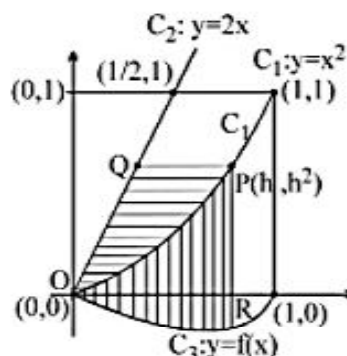
Differentiating w.r.t. h

$$\left(\sqrt{h^2} - \frac{h^2}{2} \right) \cdot 2h = h^2 - f(h)$$

$$\Rightarrow 2h^2 - h^3 = h^2 - f(h)$$

$$\Rightarrow f(h) = h^3 - h^2$$

$$\Rightarrow f(x) = x^3 - x^2$$



AREA ENCLOSED IN CASE ONE CURVE ARE EXPRESSED IN POLAR FORM :

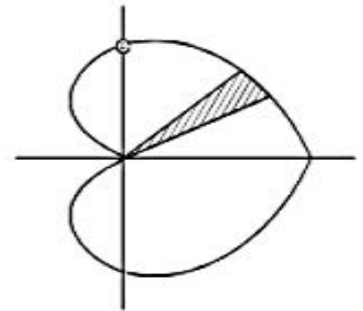
$$\text{Area of any curve} = \frac{1}{2} \int r^2 d\theta$$

Illustration :

Find the area of the cardioid $r = a(1 + \cos\theta)$

$$\text{Sol. } A = \frac{1}{2} \int_0^{2\pi} r^2 d\theta = \frac{a^2}{2} \int_0^{2\pi} 4 \cos^4 \frac{\theta}{2} d\theta \quad \text{put } \frac{\theta}{2} = t$$

$$A = a^2 \int_0^{\pi} 4 \cos^4 t dt = 8 \times \frac{3\pi a^2}{16} = \left(\frac{3\pi a^2}{2} \right)$$



AREA IN RESPECT OF CURVE REPRESENTED PARAMETRICALLY :

Illustration :

Find the area enclosed by the curves $x = a \sin^3 t$ and $y = a \cos^3 t$

$$\text{Sol. } x^{2/3} + y^{2/3} = a^{2/3}$$

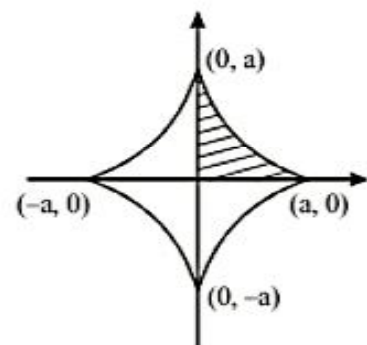
$$\text{Required area} = 4 \int_0^a \left(a^{2/3} - x^{2/3} \right)^{3/2} dx$$

$$\text{Put } x = a \sin^3 t ; dx = 3a \sin^2 t \cos t dt$$

$$\text{Area} = 4 \int_0^{\pi/2} \left(a^{2/3} - a^{2/3} \sin^2 t \right)^{3/2} 3a \sin^2 t \cos t dt$$

$$A = 12a^2 \int_0^{\pi/2} \sin^2 t \cos^4 t dt \quad \dots (1)$$

$$A = 12a^2 \int_0^{\pi/2} \sin^4 t \cos^2 t dt \quad \dots (2)$$



Adding (i) and (ii)

$$\begin{aligned}
 A &= \frac{12a^2}{2} \int_0^{\pi/2} \cos^2 t \sin^2 t (\cos^2 t + \sin^2 t) dt \\
 &= 6a^2 \int_0^{\pi/2} \sin^2 t \cos^2 t dt = 6a^2 \int_0^{\pi/2} \frac{\sin^2 2t}{4} dt = \frac{3a^2}{2} \int_0^{\pi/2} \left(\frac{1 - \cos 4t}{2} \right) dt \\
 &= \frac{3a^2}{4} \left(t - \frac{\sin 4t}{4} \right)_0^{\pi/2} = \frac{3a^2}{4} \left(\frac{\pi}{2} \right) = \frac{3\pi a^2}{8}
 \end{aligned}$$

Practice Problem

- Q.1 For what value of k is the area of the figure bounded by the curves $y = x^2 - 3$ and $y = kx + 2$ is the least. Determine the least area.
- Q.2 Find the area enclosed by the parabola $(y-2)^2 = x-1$ and the tangent to it at $(2, 3)$ & x -axis.
- Q.3 Area enclosed between the smaller arc of the circle $x^2 + y^2 - 2x + 4y - 11 = 0$ and the parabola $y = -x^2 + 2x + 1 - 2\sqrt{3}$.
- Q.4 Find the area of the figure bounded by the parabola $y = ax^2 + 12x - 14$ and the straight line $y = 9x - 32$ if the tangent drawn to the parabola at the point $x = 3$ is known to make an angle $\pi - \tan^{-1}6$ with the x -axis.
- Q.5 Find the area bounded by the curve $g(x)$, the x -axis and the ordinate at $x = -1$ and $x = 4$ where $g(x)$ is the inverse of the function $f(x) = \frac{x^3}{24} + \frac{x^2}{8} + \frac{13x}{12} + 1$
- Q.6 $f(x) = x^3 + 3x + 2$ and $g(x)$ is the inverse of it. Then compute the area bounded by $g(x)$, x -axis and the ordinate at $x = -2$ and $x = 6$.

- Q.7 Find the value of the parameter 'a' for which the area of the figure bounded by the abscissa axis, the graph of the function $y = x^3 + 3x^2 + x + a$, and the straight lines, which are parallel to the axis of ordinates and cut the abscissa axis at the point of extremum of the function, is the least.
- Q.8 Find the average value of y^2 w.r.t. x for the curve $ay = b\sqrt{a^2 - x^2}$ between $x = 0$ & $x = a$. Also find the average value of y w.r.t. x^2 for $0 \leq x \leq a$.

Answer key

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|---------------------------------------|--|--|
| Q.1 $k = 0, A = \frac{20\sqrt{5}}{3}$ | Q.2 0009 | Q.3 $4\left(\frac{8 - 3\sqrt{3} + 2\pi}{3}\right)$ |
| Q.4 $\frac{125}{2}$ | Q.5 $\frac{16}{3}$ | Q.6 $\frac{9}{2}$ |
| Q.7 $a = -1$ | Q.8 (i) $a = \frac{2b^2}{3}$, (ii) $b = \frac{2b}{3}$ | |
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